APPENDIX B: INNOVATION-BASED ADAPTIVE KALMAN FILTERING

To track the continuous movement of Eves in 3D space, we consider both their position and velocity. For the e-th Eve, we define its state vector $\mathbf{s}_e[n] \in \mathbb{R}^{6\times 1}$ at time slot n as

$$\mathbf{s}_e[n] = [x_e[n], y_e[n], z_e[n], v_{x,e}[n], v_{y,e}[n], v_{z,e}[n]]^T \tag{78}$$

where the first three elements represent the 3D position coordinates, and the remaining three represent the corresponding velocities. Assuming a baseline constant velocity (CV) model for short intervals, the state transition is modeled as a linear Markov process

$$\mathbf{s}_e[n] = \mathbf{F}\mathbf{s}_e[n-1] + \mathbf{n}_{q,e}[n] \tag{79}$$

where $\mathbf{F} \in \mathbb{R}^{6\times 6}$ is the state transition matrix, expressed as

$$\mathbf{F} = \begin{bmatrix} \mathbf{I}_3 & \delta\mathbf{I}_3 \\ \mathbf{0}_{3\times 3} & \mathbf{I}_3 \end{bmatrix} \tag{80}$$

The term $\mathbf{n}_{q,e}[n] \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}_e[n])$ is the process noise vector with a time-varying covariance matrix $\mathbf{Q}_e[n]$, reflecting the uncertainties caused by the Eve's unknown maneuvers (e.g., sudden accelerations). The observation obtained by the collaborative BS system at time slot n is the 3D coordinate $\hat{\mathbf{w}}_e^{\mathrm{LS}}[n]$ estimated via the LS method. The observation vector $\mathbf{m}_e[n] \in \mathbb{R}^{3\times 1}$ is defined as

$$\mathbf{m}_e[n] = \hat{\mathbf{w}}_e^{\mathrm{LS}}[n] = \left[\hat{x}_e^{\mathrm{LS}}[n], \hat{y}_e^{\mathrm{LS}}[n], \hat{z}_e^{\mathrm{LS}}[n]\right]^T \tag{81}$$

The observation equation extracts the position state via a linear transformation

$$\mathbf{m}_e[n] = \mathbf{H}\mathbf{s}_e[n] + \mathbf{n}_{r,e}[n] \tag{82}$$

where the observation matrix $\mathbf{H}$ is defined as $\mathbf{H} = \begin{bmatrix} \mathbf{I}_3 & \mathbf{0}_{3\times 3} \end{bmatrix}$. The observation noise $\mathbf{n}_{r,e}[n] \sim \mathcal{N}(\mathbf{0}, \mathbf{R}_e[n])$ originates from the VA-MUSIC angle estimation errors and the LS geometric dilution of precision. Its covariance matrix $\mathbf{R}_e[n]$ is essentially time-varying depending on the Eve-to-BSs distance. To prevent performance degradation caused by mismatched fixed covariances, the IAKF algorithm recursively executes the following prediction and adaptive update stages. In the prediction stage, the prior state and error covariance are computed as

$$\hat{\mathbf{s}}_e[n|n-1] = \mathbf{F}\hat{\mathbf{s}}_e[n-1|n-1] \tag{83}$$

$$\mathbf{P}_e[n|n-1] = \mathbf{F}\mathbf{P}_e[n-1|n-1]\mathbf{F}^T + \mathbf{Q}_e[n-1] \tag{84}$$

Before the update stage, we introduce the innovation sequence (residual vector), which represents the discrepancy between the actual LS measurement and the predicted observation

$$\boldsymbol{\iota}_e[n] = \mathbf{m}_e[n] - \mathbf{H}\hat{\mathbf{s}}_e[n|n-1] \tag{85}$$

To adaptively capture the dynamic tracking environment, we estimate the practical innovation covariance matrix $\hat{\mathbf{C}}_{\iota,e}[n]$ using a sliding window of size W

$$\hat{\mathbf{C}}_{\iota,e}[n] = \frac{1}{W} \sum_{j=n-W+1}^{n} \boldsymbol{\iota}_e[j] \boldsymbol{\iota}_e^T[j] \tag{86}$$

Based on the covariance matching principle, the measurement noise covariance $\mathbf{R}_e[n]$ and process noise covariance $\mathbf{Q}_e[n]$ are dynamically updated to align the theoretical statistical properties with the actual varying errors

$$\mathbf{R}_e[n] = \max \left\{ \mathbf{R}_{\min}, \hat{\mathbf{C}}_{\iota,e}[n] - \mathbf{H}\mathbf{P}_e[n|n-1]\mathbf{H}^T \right\} \tag{87}$$

$$\mathbf{Q}_e[n] = \max \left\{ \mathbf{Q}_{\min}, \mathbf{D}_e[n-1] \hat{\mathbf{C}}_{\iota,e}[n] \mathbf{D}_e^T[n-1] \right\} \tag{88}$$

where $\mathbf{R}_{\min}$ and $\mathbf{Q}_{\min}$ are predefined positive definite matrices to ensure numerical stability and prevent matrix singularity during tracking. With the adaptively updated covariances, the Kalman gain $\mathbf{D}_e[n]$ is calculated as

$$\mathbf{D}_e[n] = \mathbf{P}_e[n|n-1]\mathbf{H}^T \left(\mathbf{H}\mathbf{P}_e[n|n-1]\mathbf{H}^T + \mathbf{R}_e[n] \right)^{-1} \tag{89}$$

Finally, the posterior state estimation and the error covariance matrix are updated

$$\hat{\mathbf{s}}_e[n|n] = \hat{\mathbf{s}}_e[n|n-1] + \mathbf{D}_e[n] \boldsymbol{\iota}_e[n] \tag{90}$$

$$\mathbf{P}_e[n|n] = \left(\mathbf{I}_6 - \mathbf{D}_e[n]\mathbf{H} \right) \mathbf{P}_e[n|n-1] \tag{91}$$

By extracting the first three components of $\hat{\mathbf{s}}_e[n|n]$, we obtain the highly accurate 3D coordinates $\hat{\mathbf{w}}_e[n]$.